MSX and SPECTRAVIDEO
COMPUTER FORUM

Volume 1 No. 10

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アリババと40人の盗賊

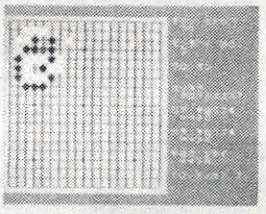
.ROM 8K 4,500円 SONY




アラビアン・ナイトの中ではアリババは、たったひとこと「ひらけゴマ！」と言っただけでお金持ちになったけれど、ＭＳＸゲームのアリババはラクにお金を稼げない。迷路の中で40人もの盗賊どもと戦わなければならないのダ。ガンバレ！

MSX実戦囲碁

3.5in. 32K 6,000円 SONY

ツケ・ハリ・キス・コスミ・オキなどの基本手筋について208個のパターンを用意。出題は「藤沢秀行囲碁教室３・囲碁基本手筋（土屋書房刊）」に準拠している。間違った問題だけを復習することだってできるから、メキメキ上達してしまう！

クイックノート

3.5in. 64K 14,800円 SONY




アドレス帳や家計簿に、野球のスコアブックやレコード管理帳に。もう使い方次第で応用範囲がどんどん広がる便利なソフト。計算やサーチ、編集プリントなどの機能もついている。データの最高処理能力は2,800！　キミは何を処理するのかな!?

α-スクアドロン

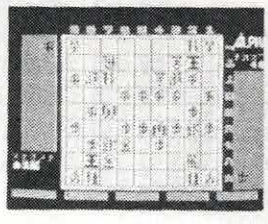
.ROM 8K 4,000円 SONY




キミはスクアドロンのコックピットに座り、地球を防衛する。そして、ブラックホールに突入し、地球を征服しようとする敵を迎え撃つ。その戦いを終え、今度は地球に帰還、着陸する。この間、少しでも操縦をミスすると墜落してしまうゾ！

MSX将棋

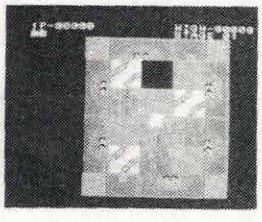
.ROM 8K 5,800円 SONY

日本男児なら、将棋ぐらいできないと恥だ。そこで、このソフト。これが、なかなかの本格派なのだ。同じ局面でキミが同じ手を指す。今までだとコンピュータも同じ手で応じるんだけど、これは違った手を指したりするんだ。何回戦っても飽きがこないゾ。

クレイジートレイン

.ROM 8K 4,500円 SONY

運転手が居眠りをしてしまったのか、それとも整備にミスがあったのか!?　とにかく乗客を満載した列車が暴走をしてしまった。さあ、一大事だ。この危機を救うには、クレイジートレインが脱線、追突しないようポイントを次々と変えるしかないのだ。

E.I.

エディー2

GAME ABC

Broderick J. Brown

PRODUCTION CONSULTANT
Paul Hammill

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Courtesy: Foxcom Pty Ltd, Fremantle.

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MSX and SPECTRAVIDEO

COMPUTER FORUM

Magazine

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Broderick J. Brown

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Paul Hammill

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EDITORIAL

MSX and SV hardware is in short supply in Australia. Software is still available and the MSX range is increasing, but there are few MSX stockists remaining.

A couple of companies are importing the latest titles. These people have 'stuck their necks out' because they must buy in large quantities - sometimes fifty pieces of each title. Most computer shops cannot boast that sort of turnover from personal shoppers. They must rely on mail orders. Country people use mail order for goods that are not available in their local town.

It is not easy deciding which piece of software to buy even if you can visit a shop and look at the range. This is where our software reviews can help. Computer Forum will continue to bring you reviews of the latest available software as long as the importers send us review copies. They will only do so if you are prepared to order something by mail order.

Your mail order safeguard is that the supplier wants repeat business. There are not enough MSX and SV owners to just make one sale to each and stay in business. If you experience difficulty with any business advertising in Computer Forum, we would like to know about it.

Happy computing ...

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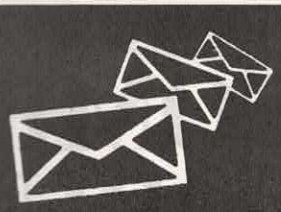
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MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

MSX & SV BBS

I am writing to advise that Spectravideo 328 is now in operation as a 'BBS' (Bulletin Board System). The BBS has six sub-boards with one (Z80 sub-board) for use of Spectravideo and MSX users to leave and receive questions and replies.

The hours of operation are 24 hours daily running at 300 baud only. The number is 075 46 3252.

Brendan Pratt
Sysop SVI-RBBS

PS I believe that this is the very first BBS on a Spectravideo in Australia - maybe the world??

Thanks for the info, Brendan, and you could be right.

256K RAM FOR SV318

I am looking at the possibility of upgrading my SV318 to 256K RAM. Impossible do I hear you all say? I wonder. I've tried 256K chips and they work to a degree. It will load a Basic program and an ASCII file but with some very funny results from the function keys. It wasn't interested in a machine code load until I connected pin 1 to +5 volts. Now the ASCII file would not load. Using machine code switching of banks, we should be able to access at least 8 x 28815 byte pages if, by programming, we can talk to address line A8 on pin 1.

I'd also very much like to use a high res RGB monitor with my 318. Do you know of a way to connect or, as I've been informed, build a small triple amplifier to output to an RGB monitor. As you might have guessed I'm crazy enough to have a go at just about anything. What do you think?

John Tillotson
Romsey, Victoria

This letter caused a few hours of midnight oil burning. Your theory about paging the RAM is sound but our concern was the refresh line and just how complex the memory management coding would be. MSX2 machines appear to use this theory with their 'memory mapper' control software. If we can find out more about that, we will certainly let you know.

Connecting an RGB monitor should be possible - the Sony MSX has an RGB outlet already. The signals you need are present on the VDP chip. They are Pin 36 (Y output with sync and luminance), Pin 38 (R-Y signal) and Pin 35 (B-Y signal) both of these with unmodulated chrominance. These can then be run to an RGB encoder/driver. The background to this is given in the Texas Instrument publication 9900 Data Manual.

While on the subject of crazy ideas, how about a 256K RAM controlled by a slave processor to act as a RAM-DISK?

JOYSTICK PORTS AND THINGS

I own a SV328 and was just about to trade it in when I found your magazine. Now, with the extra information you've given me, I am quite happy to keep it.

Could you please print a section on the joystick port. Page 165 of the 328 users' manual shows briefly what each pin is for but not in detail. Also could you explain how paddles and the touch pad connect to the port so I can simulate them.

I found that, for example, INPUT A$ does not work while in SCREEN 1 or 2 but LINE INPUT A$ does - at least until ENTER is pressed. Then the computer jumps to a mixed screen mode. I always use LINE INPUT because it doesn't have the "?" prompt and looks better in a program -
eg. 10 LINE INPUT ">";W$: ' for a string
20 LINE INPUT ">";N$:N=VAL(N$): ' for a value

I was wondering what effect it would have if Computer Forum was advertised on TV. I know some SV owners who didn't know about it and even I came across it by accident. This mag needs to be kept alive because it's a vital link in the survival of SV machines. You should ask each reader to contribute at least one thing each month and I think you'd get a good response. Thanks for the best info on SV machines. Are there any kits for SV products?

Jeffrey Dean
Nullawarre, Victoria

Thanks for the wrap. The joystick ports are a digital connection, and to use any analogue unit with them requires an analogue to digital (A/D) converter or a burst pulse unit which determines the length of the burst with the value of the potentiometer in the paddle.

We did consider TV advertising - for about ten minutes. Even with a large budget, we would need to advertise on every channel in every state and at what time? Peak time ads are booked months in advance.

Your idea of every reader contributing something is exactly the idea of Computer Forum - shared information. What about it readers? If you would like to help the magazine and other SV and MSX owners you could try placing notes on notice boards at clubs, schools, lunch rooms, delis etc. - but ask permission first. The more readers we have the more information and support you have.

MSX 80 COLUMN CARD

Could you please tell me if there is a way of running 80 columns under Disk Basic. I have an SV728 (MSX) and SV707 Disk Drive and 80 Column cartridge. CP/M is OK but have not been able to figure out how to get it to work in MSX. I have a switch on the monitor between 40 and 80 column.

G Mates
Frankston, Victoria

We have not been able to get hold of an 80 Column card to test your problem and cannot advise how you can achieve 80 Column output using Disk Basic. Can any of our readers help?

SV REBOOT

This causes total boot-up including Spectravideo logo and Screen 0 and Basic version.

```
0 REM FOR SVI318 & 328
10 DEFUSR0=0
20 PRINT USR(0)
30 END
```

Please keep us up to date on the MSX adaptor for SV328 & 318 as I own a 318 MKII. Could you also ask for the new address of Mountain Valley Software as I sent to them asking for a catalogue. Unfortunately they have moved. The old address was PO Box 407, Boronia, Vic.

Craig Hursey
Stanley, Tasmania

Thanks for the tips Craig. Unfortunately, the MSX adaptor is no longer being manufactured. We have heard of a piece of software which allows the SV to read some MSX tapes and are trying to get a copy.

I don't know what happened to Mountain Valley Software either, can anyone help?

PROGRAMMING HELP

When sprites overlap the ONSPRITEGOTO can replace one or both with pictorial explosions. How d'ya get rid of them? My solution was to put an invisible SPRITE (color 0), but this does not seem satisfactory. Another sprite could overlap the invisible one.

Is an inaudible SOUND used to stop a sound effect going on forever? Finally, my Sony HB-75AS has two cartridge slots. In what circumstances can they both be used at the same time? Can I put a 4K RAM Data Cartridge in one slot and a program cartridge in the other or is the second slot for an expansion unit not available in Australia? Pray tell.

Gordon Browell
Biggenden, QLD

The way to get rid of sprites that have collided is to place them off screen by using 208 or 209 for their vertical location.

To stop a sound, use either SOUND7,255 or SOUNDX,0 where X is 8,9 or 10 for channel A,B or C. The top slot on your Hit Bit is generally known as the cart ridge slot and the rear one is the expansion slot, although you can plug in a cartridge. The philosophy being to plug hardware (eg disk controller) in the back and firmware (cartridge software) in the top. You can plug a 4K RAM cartridge in one slot and a program in the other but the program would need to recognise the existence of the RAM cartridge.

WES NICHOLSON PRESENTS

THE CATACOMBS OF DOOM

for your entertainment. A text adventure with a difference for CP/M and PC compatibles.

Costs only $12 for the game, a user guide, set of map sheets and postage. Ten sets of map sheets also available for $2.60 including postage.

Available only from me at -

PO BOX E271, QUEEN VICTORIA TERRACE, ACT 2600

CP/M users: You must have, or emulate, a VT-52 or ADM monitor. Please specify which when ordering.

Overseas: Please add $3 for airmail.

Software NEWS & REVIEWS

METEOR SWARM
(SV318/328 Cassette)
by Electric Adventures



In this machine-code game, you control a space ship which is trapped in a meteor belt and must blast your way out. You rotate left or right and push forward for thrust - pulling back on the joystick moves you to hyperspace. This can get you out of a sticky position but you might come out of hyperspace right in the path of a meteor. Other ships are also caught in the belt and will shoot at you if you don't destroy them first.

Also included is a bonus game called Birds of Orion, a very difficult invaders type game.

Publisher: Forrest Data Services
Price: $19.95

THE CATACOMBS OF DOOM
(SV Disk)

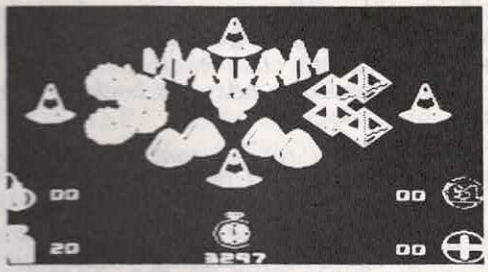
This is a CP/M Text Adventure available on disk for Spectravideo. The object of the adventure is to move through the rooms of the catacombs collecting all kinds of treasure, food, weapons and potions. These will help you fight the monsters you encounter and survive this long ordeal. You can also move up and down to different levels.

To control your adventurer, you enter a verb/noun combination and must eat and drink regularly to maintain your fighting level. At any time, you can save the game to be continued later - a handy feature in an adventure with approximately 300 rooms (the number of rooms changes each time you start a game).

Documentation for a game like this is essential and the enclosed instructions are very clear and helpful. THE CATACOMBS OF DOOM comes on disk with instruction manual and a set of map sheets. This one is VERY hard!

Publisher: Wes Nicholson
Price: $12.00 posted

MOLECULE MAN
(64K MSX Cassette)



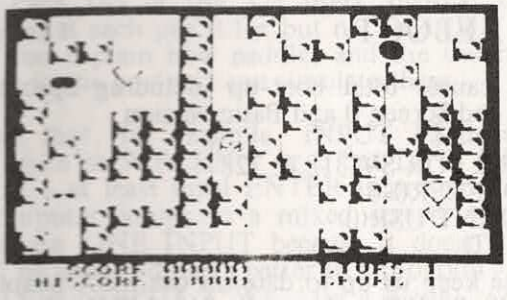
As the Molecule Man, your mission is to live and escape from the gigantic 3D maze in which you are trapped. The radiation is making you progressively weaker, and you must find coins and buy life pills. You may also buy bombs in order to get to the 16 circuits you need to operate the teleporter which will get you off this perplexing planet.

1 player, no music, excellent graphics, great concept
- good value.

Publisher: Mastertronic
Price: $9.95

Review copy supplied by LPG Software

ICE (64K MSX Cassette)

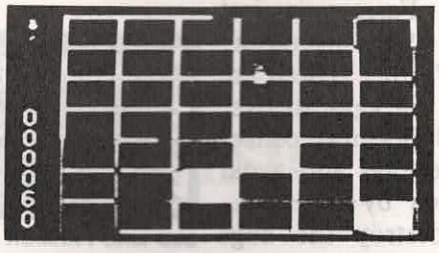


You are a penguin, surrounded by ice. Your objective is to get through to the real ice, diamonds, which are scattered around you. You must push the diamonds together, so that they are all touching. This sounds pretty easy, but you also have to avoid three rather nasty looking ice-monsters. Using the keyboard or a joystick, you can move around between the ice blocks and push them around to trap the monsters. You need good tactics and quick fingers for this one. Good luck!

Publisher: Eaglesoft
Price: $14.95

Review copy supplied by LPG Software

PANEL PANIC (64K MSX Cassette)



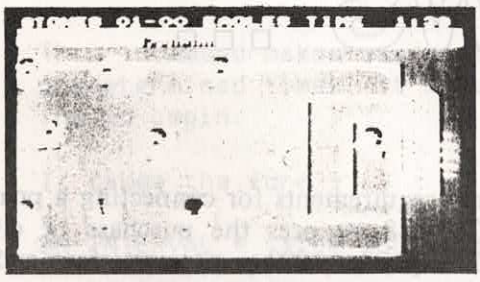
You control the paint brush and must trace the outlines of a screen full of squares in order to paint them. Your objective is to paint the entire screen red and to escape the monsters chasing you. Set your brush to work.

An original game for 1 player, average music, reasonable graphics, 1 screen, 4 levels.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software

FOOTBALL (MSX Cartridge)



This is a game of soccer (English football). You are the captain of your local soccer club. Today is the day of the big match against the best team in the country, and you must win to regain the honour of being on top! Move your men with the joystick or keyboard and pass, dive, shoot and save your way to victory.

You choose the team names and colours, and its up to you to determine when half-time will be. Go get 'em!

1 or 2 players, good graphics and music - it scores!

Publisher: Konami
Price: $49.95

Review copy supplied by LPG Software

MOONRIDER (64K MSX Cassette)



You are out on patrol in your Moon Rider and must reach your destination as quickly as possible. To do this, you must battle to scramble your way over the rough terrain, while under fire from the planet's natives. It's hair-raising action, so good luck! You may need it!

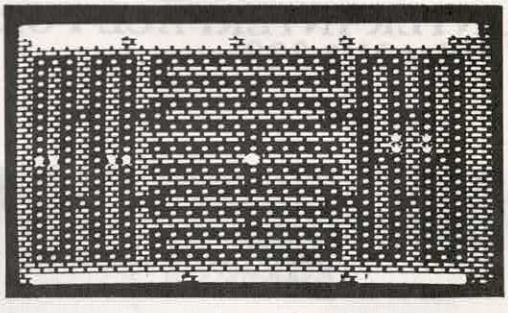
Similar to the Moon Patrol arcade game and uses keyboard or joystick. It takes a little while to get used to pushing the joystick forward to jump - the trigger is used to fire your weapons.

1 player, good music, speech and graphics - we couldn't stop playing this one.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software

MAZES UNLIMITED (64K MSX Cassette)



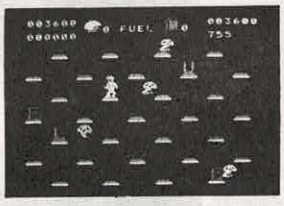
Search your way through dozens of mazes, chewing up power pills as you go. Your aim is to clear the screen of pills, but watch out! You are relentlessly pursued by four rather nasty characters of ill intent. Fire at them to send them back to the corner they came from - but don't be too trigger happy! You only have twenty bullets.

An updated Pac-Man game where you can set your speed and that of your pursuers. 1 player, little music, good graphics, 16 screens.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software

TETRA HORROR (SV318/328 CASSETTE)


TETRA HORROR SD216T

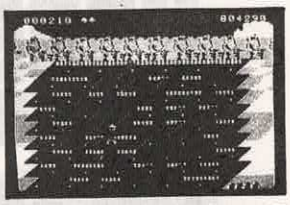
Tano, the adventurer, has just found a map of an ancient city in South America. There is treasure there and he must collect it. To begin his adventure, Tano collects fuel to power his fire gun. He then must enter Tetra Castle, where he has to destroy many skulls to get to the diamond treasure awaiting him. He must proceed carefully because one false step and he plunges into further peril! Indiana Jones eat your heart out!

1 or 2 players, average graphics, 3 screens, 3 levels - good concept.

Requires extra memory for use with SV318

Publisher: Spectravideo
Price: $19.00

OLD MAC FARMER (SV318/328 Cassette)


OLD MAC FARMER SD211T

Old Mac Farmer is just about to plant his crops, but what's this! There are bugs and big eggs all over his fields! He must collect up all the eggs and avoid being eaten by the bugs. He can destroy the bugs with insecticide spray, but he only has a limited supply. What an effort! He's cleared his fields of bug eggs, now he must plant his seeds. Oh no! He is being plagued by birds! Will this never end? Poor Old Mac Farmer.

1 or 2 players, good graphics and music. Pacman style game which is much more interesting than the original. 1 screen, 2 levels.

Requires extra memory for use with SV318

Publisher: Spectravideo
Price: $19.00

PRODUCT NEWS ...

PRINTER INTERFACE FOR SV318 AND 328



The normal requirements for connecting a printer to your SV318/328 requires the purchase of either a 601 or 602 expander, 802 printer interface and cable or an SV605 expander and cable.

A cheaper and simpler solution is the SP101 Parallel Printer Interface from Forrest Data Services. This is a stand-alone interface which requires no separate power supply.

The interface simply plugs straight in the back of the SV318/328 keyboard. The printer cable then attaches between the SP101 and your printer. Supports most Centronics type printers including Epson, Star, Citizen, Seiko, NEC, and printers named variously GP80, GP100, CPA80, CPA100 etc.

Manufactured in Australia and comes with a full ninety day warranty, the SP101 retails for $84. A one metre printer cable is also available at $35.

BASIC TUTORIAL - Part 10

The TIME command

This is not actually a clock function. It uses the value of the internal system timer memory location which is incremented by 1 every sixty seconds if interrupts are not disabled - for example when doing cassette input or output.

```
10 CLS : TIME=0
20 LOCATE 10,10
30 PRINT INT(TIME/60)
40 GOTO 20
```

This program will change the number on the screen approximately once every second.

The ON INTERVAL command

This command makes use of the internal timer to perform a function at predetermined times. It requires that INTERVAL ON be executed for checking to begin.

It takes the form -

ON INTERVAL = value GOSUB line number

where value divided by 60 gives the delay in seconds between the interval occuring. Line number is the line number of the subroutine which is to be used at each interval.

```
10 ON INTERVAL = 60 GOSUB 200
20 INTERVAL ON
30 PRINT N
40 N=N+1
50 GOTO 30
200 BEEP
210 RETURN
```

The PRINT USING command

This command is used to tidy up or align the displayed values. Without this command, all printing will start at the current cursor position and

would look very messy particularly with decimal amounts -

```
Example:  10 A=1234:B=56.7:C=1.956
          20 PRINT A
          30 PRINT B
          40 PRINT C

Displays       1234
               56.7
               1.956
```

With PRINT USING, we can determine where the numbers will be displayed in relation to a predetermined format.

(Hash symbol) This is used to indicate how many number positions are used in the print field. If there are insufficient numbers to fill the field then spaces are added at the start -

```
Example:  10 A=1234:B=76
          20 PRINT USING "####";A
          30 PRINT USING "####";B

          Displays       1234
                           76
```

To overcome having to use the exact format each time a PRINT USING command is given, the format string can be defined first as a variable -

```
Example:  10 A=1234:B=76
          20 F$="####"
          30 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays       1234
                           76
```

A decimal point can be included anywhere in the string of hashes and any numbers exceeding the trailing locations will be rounded or if any space is left then zeroes will be added -

```
Example:  10 A=12.346:B=7.6
          20 F$="##.##"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays       12.35
                          7.60
```

+ (Plus sign) This can be used at the start or end of a format string to indicate that the sign of the number, either plus or minus is to be

printed at that point -

```
Example:  10 A=12.346:B=-7.6
          20 F$="+##.##"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays        +12.35
                           -7.60

Example:  10 A=12.346:B=-7.6
          20 F$="##.##+"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays        12.35+
                           7.60-
```

- (Minus) Using a minus sign at the end of a format string will cause only the negative numbers to have a minus sign printed after them -

```
Example:  10 A=12.346:B=-7.6
          20 F$="##.##-"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays        12.35
                           7.60-
```

** (2 asterisks) Two asterisks at the start of a format string indicate that all leading spaces are to be filled with this sign. The two asterisks also count as two print positions -

```
Example:  10 A=12.346:B=7.6
          20 F$="**##.##"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays        **12.35
                          ***7.60
```

$$ (2 dollar signs) These cause a dollar sign to be precede the amount when displayed. The two signs also indicate two print positions one of which will be the sign itself -

```
Example:  10 A=12.346:B=7.6
          20 F$="$$##.##"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays         $12.35
                            $7.60
```

**$ (2 asterisks and dollar) This is a combination of the last two and results in a dollar sign in front of the amount and the leading spaces to be filled with asterisks -

```
Example:  10 A=12.346:B=7.6
          20 F$="$$###.##"
          20 PRINT USING F$;A
          30 PRINT USING F$;B

          Displays        **$12.35
                          **$7.60
```

, (Comma) If this is used in front of a decimal point, then a comma will separate every group of three digits to the left of the decimal -

```
Example:  10 A=123456789
          20 F$="##########,.##"
          20 PRINT USING F$;A

          Displays        123,456,789
```

A comma placed at the end of a format string will cause the numbers printed to be separated by a comma -

```
Example:  10 A=235:B=17:C=5
          20 F$="###.##,"
          30 PRINT USING F$;A;B;C

          Displays        235.00, 17.00,  5.00
```

^^^^ (4 Carats) If these are used at the end of a format string, then the number will be displayed in exponential format -

```
Example:  10 A=.00021
          20 PRINT USING "##.##^^^^";A

          Displays        2.10E-04
```

If the number to be displayed is too large for the format string, the the percentage sign (%) will be displayed to the left of the number -

```
Example:  10 A=1234
          20 PRINT USING "###";A

          Displays        %1234
```

PRINT USING can also be used with string variables -

! (Exclamation mark) This indicates that only the first letter of the string is to be displayed -

```
Example:  10 A$="ABC":B$="DEF":C$="GHI"
          20 PRINT USING "!";A$
          30 PRINT USING "!";B$
          40 PRINT USING "!";C$

Displays      A
              D
              G
```

\\ (2 backslashes) These require that the string be displayed for the length of the number of spaces between the backslashes plus two for the symbols -

```
Example:  10 A$="ABCDEF"
          20 PRINT USING "\    \" : REM four spaces between symbols
          30 PRINT USING "\  \"   : REM two spaces between symbols

          Displays      ABCDEFG
                        ABCD
```

The LPRINT USING command

This command functions in exactly the same way as the PRINT USING command except that output will be to a printer if one is attached. This is very convenient, because you can format the type of output on the screen with PRINT USING then add L to the start of PRINT and the printout will appear the same - saves a lot of paper.

MATHS COMMANDS

The ABS command

This gives the ABSolute value of the argument contained in the brackets following the command.

```
Example:      100 X=123                  100 PRINT ABS(-123)
              110 PRINT ABS(X)

              Displays 123               Displays 123
```

The ATN command

The ArcTaNgent (in radians) gives the angle whose Tangent is contained in brackets following the command

```
Example:    100 X=123
            110 PRINT ATN(X)

            Displays 1.56267
```

To obtain degrees, multiply the result by 57.29578

The CINT command

Converts a value to a whole number. Works in a similar fashion to the INT command.

```
Example:    10 INPUT "ENTER DECIMAL";N
            20 PRINT CINT(N)

            enter 1.2345678912345

            Displays 1
```

The CSNG command

Converts a value to single-precision decimal.

```
Example:    10 INPUT"ENTER NUMBER";N
            20 PRINT CSNG(N)

            enter 1.2345678912345

            Displays 1.23457
```

The CDBL command

Converts a value to double-precision decimal.

```
Example:    10 DEFINT A-Z
            20 A=13 : B=12
            30 C=A/B
            40 PRINT C
            50 PRINT CDBL(A/B)

            Displays    1
                        1.0833333333333
```

The COS command

COS(X) gives the COSine of X where X is in radians. Where X is in degrees, use COS(X * .01745329)

The EXP command

EXP(X) gives the 'natural exponential' of X, that is e to the power X. This is the inverse of the LOG function, so X=EXP(LOG(X))

The INT command

INT(X) gives the largest whole number (integer) which is not greater than the value of X.

```
Example:    100 X=123.45
            110 PRINT INT(X)

            Displays 123
```

The FIX command

Gives the whole number value of a number. When using positive numbers, FIX performs the same as INT. When negative numbers are used, FIX gives the whole number only.

```
FIX(1.2)=1          INT(1.2)=1
FIX(-1.2)=1         INT(-1.2)=2
```

The LOG command

LOG(X) gives the logarithm of X.

This is the inverse of the EXP function so that X=LOG(EXP(X))

The MOD command

This is the MODulus function and gives the remainder of a division.

```
Example:    10 PRINT 10 MOD 4
            20 PRINT 20 MOD 3
            30 PRINT 11 MOD 2.5
```

```
Displays  2    that is 10/4   has a remainder of 2
          2    that is 20/3   has a remainder of 2
          1    that is 11/2.5 has a remainder of 1
```

The SGN command

```
SGN(X) gives the value of the sign of X
If X is negative then SGN(X) = -1
If X is zero     then SGN(X) = 0
If X is positive then SGN(X) = 1
```

The SIN command

```
SIN(X) gives the sine of X if X is in radians.
If X is in degrees, use SIN(X * .01745329)
```

The SQR command

```
SQR(X) gives the square root of X

Example:        100 PRINT SQR(9)

                Displays 3
```

The TAN command

```
TAN(X) gives the tangent of X where X is in radians.
If X is in degrees, use TAN(X * .01745329)
```

LOGICAL FUNCTIONS

The AND function

```
This is the logical AND of two or more values -

Example:        10 A=2:B=3:C=4
                20 IF A<B AND B>C THEN PRINT "WRONG"
                30 IF A<B AND B<C THEN PRINT "RIGHT"

                Displays  RIGHT
```

because it checks to see if A is less than B and then checks if B is less than C

The OR function

This is the logical OR of two or more values -

```
Example:        10 A=2:B=3:C=4
                20 IF A<B OR B>C THEN PRINT "RIGHT"

                   Displays  RIGHT
```

because it checks to see if A is less than B. If A was greater then B then it would also check if B was greater than C.

DEFINING VARIABLE TYPES

Variables are assigned their type by placing a symbol after the variable name. Unless otherwise told, your computer assumes that numeric variables are double precision.

```
A  means double precision variable
A# means double precision variable
A! means single precision variable
A% means integer variable
A$ means string variable
```

The DEFINT command

Defines variables as being INTegers or whole numbers.

```
Example:  10 DEFINT A-G
```

This means that all variables A to G in the program are to be treated as INTEGER unless otherwise defined.

```
Example:  10 DEFINT A-G
          20 A$="ABC"              : 'Variable A is a string
          30 B!=1.2345             : 'Variable B is single precision
          40 C#=1.2345678912345    : 'Variable C is double precision
          50 D=E+F                 : 'D, E & F are integers
          60 K=D/7                 : 'K is double precision
```

The DEFSNG command

Defines variables as being SiNGle precision decimals.

```
Example:  10 DEFSNG D-H
```

This means that all variables D to H in the program are to be treated as SINGLE PRECISION unless otherwise defined.

```
Example:  10 DEFSNG A-G
          20 A$="ABC"              : 'Variable A is a string
          30 B!=1.2345             : 'Variable B is single precision
          40 C#=1.2345678912345 : 'Variable C is double precision
          50 D=E+F                 : 'D, E & F are single precision
          60 K=D/7                 : 'K is double precision
```

The DEFDBL command

Defines variables as being DouBLe precision decimals.

```
Example:  10 DEFDBL F-M
```

This means that all variables F to M in the program are to be treated as DOUBLE PRECISION unless otherwise defined.

```
Example:  10 DEFDBL A-G
          20 A$="ABC"              : 'Variable A is a string
          30 B!=1.2345             : 'Variable B is single precision
          40 C#=1.2345678912345 : 'Variable C is double precision
          50 D=E+F                 : 'D, E & F are double precision
          60 K=D/7                 : 'K is double precision
```

The DEFSTR command

Defines variables as being STRings.

```
Example:  10 DEFSTR A-Z
```

This means that all variables A to Z in the program are to be treated as STRINGS unless otherwise defined.

```
Example:  10 DEFSNG A-G
          20 A="ABC"               : 'Variable A is a string
          30 B!=1.2345             : 'Variable B is single precision
          40 C#=1.2345678912345 : 'Variable C is double precision
          50 D="DEF"               : 'Variable D is a string
          60 K=A+D                 : 'Variable K is a string
```

PROGRAMMING TECHNIQUES

MAKING THE MOST OF VARIABLES

As mentioned before, variables are the characters we use to signify a

number or phrase. The variable name can start with any letter of the alphabet and be up to two characters long -

A	A$	A%
AG	AG$	AG%
D1	D1$	D1%

are all valid variable names.

The exceptions are

```
IF   this is a Basic command
FN   this is a function command
TO   this is a Basic command
ON   this is a Basic command
OR   this is a Basic command
```

GARBAGE COLLECTION

In a Basic program, memory is used up by assigning values to variables. This memory space is not recovered when the value of a variable changes.

For example -

```
10 INPUT "NAME";A$
20 INPUT "NAME";A$
```

When this program is run, if we enter PHYLLIS in response to the first request, that name is stored in memory, and the variable A$ remembers where. When the program asks for a name in line 20, we enter ROGER. A$ is now equal to ROGER and the variable A$ remembers where in memory he is stored.

But what about PHYLLIS? She is still stored in memory, but A$ has had to forget where because it can only remember one thing at a time - like a few people I know.

Garbage collection is what the computer does when it starts running out of memory space. If a program uses a very large amount of information which is constantly changing, you will notice a few pauses in the program where the computer appears to be 'frozen'. This is when it is off collecting up all of the unwanted strings and values to recover memory space. Poor old PHYLLIS is thrown out with the garbage.

This garbage collection can occur in the middle of an important part of the program. To arrange for it to happen at a less noticeable time we can force our own garbage collection using -

```
X=FRE(0)
```

ANSWERS to the questions in issue 9

We do not normally supply answers to last issue's questions, but thought we might explain why some of those programs were wrong.

1. Return without GOSUB
   This is because the program allows execution to go crashing into line 500 without being sent there by a GOSUB. To overcome this we could use line 130 END

2. If there has been no previous DIM statement for the array then the error message 'Subscript out of range' will be displayed.

3. Unidentified line number because there is no line 120

QUESTIONS

1. SCREEN 0 is a

   (a) Graphics screen
   (b) Text screen

2. What is wrong with this?

```
10 POKE 4000,255
```

3. What will be displayed after running this?

```
10 A=56:B=24
20 SWAP A,B
30 PRINT A
```

4. What will happen when this program is run?

```
10 PRINT "ABCDE"
20 PRINT "12345"
30 LIST 20
```

5. Will this cause a 'type mismatch' error?

```
10 DEFINT A-Z
20 A="ABC"
30 PRINT A
```

HINTS & TIPS

Max Cowin provides these two very useful utilities.

The first, KEYS, is a simple program for changing the function keys to what you may need. In line 30 I toggle into uppercase and turn the caps-lock light on. It is very basic and easy to follow, and can be set to run at power up as part of an Autoexec file.

```
10 WIDTH40:COLOR 15,4,4:CLS
20 '
30 POKE &HFCAB,255:OUT &HAA,0    ' SETS T
HE KEYBOARD TO UPPERCASE AND TURNS
40 'THE CAPS-LOCK LIGHT ON
50 '
60 ' LOAD THE KEYS
70 KEY1,"FILES"+CHR$(13)
80 KEY2,"LOAD"+CHR$(34)
90 KEY3,"SAVE"+CHR$(34)
100 KEY4,"LIST "
110 KEY5," RUN "
120 KEY6,"COLOR 15,4,4 "+CHR$(13)
130 KEY7,CHR$(&H5F)+"SYSTEM"+CHR$(13)
140 KEY8,"BASIC"+CHR$(13)
150 KEY9,CHR$(63)+"DSKF"+CHR$(40)+CHR$(4
8)+CHR$(41)+CHR$(13)
160 KEY10,"AUTO"
170 '
180 NEW  '  READY FOR NEXT PROGRAM
```

The second program, LOOK4IT, finds the start, end and operating address of machine code programs. To save space and make it as small as possible, there are no REM statements. To cut it down further, lines 20, 30, 80, 100 and 160 can be left out.

Line 10 sets the variables used.
Lines 30 to 70 clear some memory and poke the capital letter A into memory from 8500Hex up to DF86Hex. I use these addresses for two reasons - I found 8500H to be about as low as you an go and still have room for this program. The second address is just below the area set aside for part of Disk Basic. The Yamaha disk has a lower ceiling due to the fact that it's a Double Sided Double Density (DSDD) disk drive and needs a bit more room than Single Sided Double Density (SSDD) drives.

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Lines 80 and 90 Bload the machine code program without running it from the cassette.
Lines 100 to 140 start at the lower address and looks (PEEK's) up until the capital letter A is NOT found. The address of the first NOT A is displayed and then jumps to the next part of the search. Line 150 is there in case the program did not Bload from tape and jumps to the end of the program. There is not much use looking down as you have already gone through all of the memory.
Lines 160 to 200 start at the top and look (Peek) down for the first NOT A. The address is displayed and you pass on to the last part.
Line 210 displays the Operate address on screen.

All that is required to save the program on disk, is -
BSAVE "FILENAME.EXT", START ADDR, END ADDR, OPERATE ADDR for example -
BSAVE "WARGAMES.BIN", &H9000, &HD3FF, &HA000 where &H9000 is the start, &HD3FF is the end and &HA000 is the operate address.

As we are all aware, copying software for any purpose other than making a backup copy, is against the law. YOU HAVE BEEN WARNED! (ED)

This program is a basic one and does not work on all machine code programs. A lot of them are too big and work in the area set aside for disk basic. Happy computing fellow mushrooms!

```
10  A=0:B=0:X=0:Y=0:Z=0
20 CLS
30 PRINT "POKEING"
40 CLEAR 10,&H8500
50 FOR X=&H8500 TO &HDF86
60 POKE X,65
70 NEXT X
80 PRINT"BLOADING"
90 BLOAD"CAS:"
100 PRINT "PEEKING UP"
110 FOR Y=&H8500 TO &HDF86
120 A=PEEK(Y)
130 IF A<>65 THEN PRINT HEX$(Y):GOTO 160
140 NEXT Y
150 IF HEX$(Y)=&HDF87 THEN GOTO 220
160 PRINT "PEEKING DOWN"
170 FOR Z=&HDF86 TO &H8500 STEP-1
180 B=PEEK(Z)
190 IF B<>65 THEN PRINT HEX$(Z):GOTO 210
200 NEXT Z
210 PRINT HEX$(PEEK(&HFCC0));HEX$(PEEK(&
HFCBF))
220              END
```

HACKERS CORNER

SV ROM DECODED

Spectravideo SV-318/328 ROM version 1.1

```
6273  PRINT USING routine
6474  Linefeed
6484  Adjust printer position
648F  Adjust screen sursor posn
64F3  INKEY$
6557  NEW
657A  RUN line number
66AE  RESTORE
66C8  STOP
66CF  END
671B  CONT
672F  TRON
6730  TROFF
6735  SWAP
676E  ERASE
67A6  CLEAR
6821  NEXT
68FA  OCT$
68FF  HEX$
6904  BIN$
6909  STR$
693A  (") QUOTES
697D  Print message
6AD5  Get string
6B04  LEN
6B10  ASC
6B20  CHR$
6B2E  STRING$
6B4D  SPACE$
6B66  LEFT$
6B96  RIGHT$
6B9F  MID$
6BC0  VAL
6BF0  INSTR
6CF7  FRE
6D13  Inputline
6D9C  ENTER key
6DF5  ^C Break
6E10  ^R insert/replace
6E81  ^H backspace
6ED1  ^U clear logical line
6EDC  ^E truncate line
6EFA  ^N cursor to end of line
6F1B  ^F cursor to next word
6F2E  ^B cursor to previous word
7080  OPEN
7121  LOAD
7122  MERGE
7167  SAVE
```

MSX ROM CALLS

All MSX ROM calls should be made through the BIOS entry table.

MSX1 BIOS ENTRY TABLE

```
00BAH  Name:      GSPSIZ
       Function:  Get current sprite size
       Entry:     None
       Returns:   Sprite size (no of bytes) in A
                  Carry set if 16x16 else reset
       Modifies:  AF

00BDH  Name:      GRPPRT
       Function:  Print char on graphics screen
       Entry:     Character code in A
       Returns:   None
       Modifies:  None

0090H  Name:      GICINI
       Function:  Init PSG and data for PLAY
       Entry:     None
       Returns:   None
       Modifies:  All

0093H  Name:      WRTPSG
       Function:  Write to PSG register
       Entry:     PSG Reg in A, data in E
       Returns:   None
       Modifies:  None

0096H  Name:      RDPSG
       Function:  Read PSG register
       Entry:     PSG reg in A
       Returns:   Data in A
       Modifies:  None

00B4H  Name:      QINLIN
       Function:  Prints ? and jumps to INLIN
       Entry:     Same as PINLIN
       Returns:   Same as PINLIN
       Modifies:  All

00B7H  Name:      BREAKX
       Function:  Checks fo CTRL/STOP
       Entry:     None
       Returns:   Carry set if pressed
       Modifies:  AF

00BAH  Name:      ISCNTC
       Function:  Checks for SHIFT/STOP
       Entry:     None
       Returns:   None
       Modifies:  None
```

COMPUTER KIDS

Hi kids! Here we are again, in yet another issue of Computer Forum. This time, you get a super-sized Wordsearch puzzle to keep you busy until next issue. The words in this puzzle are all commands which you can use on your computer. If you don't know what some of them are, look them up in your computer manual. It should give a pretty good explanation. Good hunting!

Bye for now, Wendy.

COMPUWORDS

P N R U T E R O T S E R A E L D
A U I L A L R I N O T N L F I A
I N Q U D P N T N O Z C D M H T
N C T A D P D H S S R N T P A A
T O O N U E W E D I E N T E R R
G L U T M A B N C L C R G R E A
C O Y V R P U T S P R I T E A E
S R T D I R J R U A E G N M D L
N L N O L I L E T T E K P T I C
E E R E P N B D E T T I F U R B
E S W T X T R L Y F A T S Y U E
R T Q A C T E I A S C B T S K N
C E C S W D A S L C O F O R M I
S P R E S E T T P T L G P L V L

AUTO	GOTO	PRINT
CIRCLE	IF	PSET
CLEAR	INPUT	PUTSPRITE
CLOAD	INSERT	READ
CLS	INT	REM
COLOR	LEN	RESTORE
CTRL	LET	RETURN
DATA	LINE	RND
DELETE	LIST	RUN
DIM	LOCATE	SCREEN
DRAW	NEW	SOUND
END	NEXT	STEP
ENTER	PAINT	STOP
FOR	PLAY	TAB
GOSUB	PRESET	THEN

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UNDERSTANDING MACHINE CODE

To continue with this subject, it must be mentioned again that we do not intend to teach Machine Code Programming. These articles are designed to give an introduction to the subject, or to decide whether you feel you have what it takes to study the subject.

As mentioned in the last instalment, to use machine code requires a knowledge of binary maths and logic. The two are really inseparable.

INTRODUCING BINARY MATHS

Binary maths is based on zero and one, yes or no, on or off. With the decimal system, we have the units column on the right, next to that is the tens column followed by the hundreds and thousands columns. If the number in the units column exceeds nine then we add one to the tens column and if that overflows we add one to the hundreds column and so on. Adding together these two numbers illustrates just that.

```
  123
+ 277
= 400
```

The columns in binary maths have a doubling value starting with the rightmost column which is one.

BIT No.	7	6	5	4	3	2	1	0
Decimal value	128	64	32	16	8	4	2	1

As we are dealing with eight-bit computers we do not need any more. Therefore, a one in any column adds the decimal value of that column to the value of the binary number -

128	64	32	16	8	4	2	1		
0	0	0	0	0	0	0	0	=	0
0	0	0	0	0	0	0	1	=	1
0	0	0	0	0	0	1	0	=	2
0	0	0	0	0	0	1	1	=	3
0	0	0	0	0	1	0	0	=	4
0	0	0	0	0	1	0	1	=	5

and so on.

ADDING BINARY NUMBERS

Let us add together the binary equivalents of 2 and 3 decimal.

```
0010 = 2
0011 = 3
```

taking the rightmost column, add the one to the zero and place the result in that column.

```
0010 = 2
0011 = 3
   1
```

move to the next column and add the one to the one. Because the largest digit in the binary system is one, then we cannot put the result, two, in that column. So we put a zero there and carry a one.

```
0010 = 2
0011 = 3
 101 = 5
```

This can be checked on your computer with this program

```
10 A=&B0010
20 B=&B0011
30 PRINT A+B
40 PRINT BIN$(A+B)
```

BINARY MATHS AND DIGITAL ELECTRONICS

Digital electronics, as opposed to other forms of electronics, is the type used in your computer. Your radio uses electronics to produce the sound, but this is not digital. Digital electronics uses combinations of signals which are either on or off and this lends itself directly to binary maths. The volume control on your radio is not digital. It has varying degrees of ON (volume). The switch on the power point is digital, it is either ON or OFF.

Written any programs? Send them in to COMPUTER FORUM for inclusion in our Program Listings. If we decide to publish, you will receive payment based on size, originality and scope of interest. Please send programs in cassette form or disk and clearly marked which format they are for. If you wish them to be returned, enclose sufficient postage for safe return. Mark your name and address on everything.

BOOKS • BOOKS

Z80 ASSEMBLY LANGUAGE PROGRAMMING
by Lance A. Leventhal

This book assumes that you are familiar with An Introduction To Microcomputers Volume 1.

This is not just a reference book but also provides comprehensive coverage of Z80 assembly language. It contains more than 80 sample programming problems with solutions in source and object code.

Although containing the usual pages of Z80 instruction code, this book is the only one of its type that I have seen which directly relates machine code to hardware functions with circuit diagrams to aid explanation.

Not really for the beginner, Z80 Assembly Language Programming should be on the bookshelf of any student of machine code. Review copy supplied by McGraw-Hill. Publisher:

Osborne/McGraw-Hill
609 pages
Price $41.95

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

STRAWBERRY CATCHER
by Robert Hall

This listing is in SV318/328 format. For MSX owners, see note at the end of the listing.

```
1 ' ************************************
2 ' *      STRAWBERRY CATCHER          *
3 ' *              by                  *
4 ' *          ROBERT HALL             *
5 ' *            6/6/86                *
6 ' ************************************
7 SCREEN 2:COLOR 10,4,3:LOCATE13,40:PRIN
T"Strawberry              Catcher":FOR T=
1 TO 1000:NEXT:COLOR 15,4,5:SCREEN 0,0
8 WIDTH 40:CLS:LOCATE 3,8:INPUT "Do you
want instructions (y/n)";A$:IF A$="y" OR
 A$="Y" THEN GOSUB 116
9 ' ***      Title screen      ***
10 ' *** and set up variables ***
11 CLEAR 500:SCREEN 2
12 N=RND(-TIME)
13 COLOR 10,4,3:LOCATE 13,40:PRINT"Straw
berry              Catcher"
14 PLAY"o3t255120ccddeeffefedefefgfgedce
degeddedccccdcedgefegffafedefeddcdccccc"
15 PLAY "o3t255120ddeeffefgfedefedcdedcd
efefededccccccccc","o6t255120ddeeffefgfe
defedcdedcdefefededccccccccc"
16 RESTORE
17 X=128:Y=96:STR=1:B1=115:RN=50:V=10:ST
R=1:RN=RN-3:Y4=0:CY=0:C3=0:B=0:TI=0:HE=4
0:X1=0:Y1=0:SC=0:CQ=0
18 ' *** Set up hi-res screen and ***
19 ' ***      define sprites       ***
20 ON SPRITE GOSUB 80:SPRITE ON
21 SCREEN 1,2:COLOR15,4,3:CU=50:C=0:TI=0
```

```
22 FOR Y=55 TO 196 STEP 2:LINE (0,Y)-(25
6,Y),12:C=C+1:Y=Y+C:NEXT:LINE (0,55)-(25
6,55),7:CIRCLE (250,0),30,10,3.14,6.28:P
AINT(250,0),10:LOCATE 222,9:COLOR1:PRINT
"  SUN"
23 PSET (45,179):DRAW "H5U50R20D50G5L10U
50R10D50U50E5L20F5H5D50R20BR70BU70D25R25
U25L25E7R25G7E7D25G7L25E7R25L25U25"
24 LOCATE 132,112:PRINT "CUBE":LOCATE48,
134:PRINT "T":LOCATE 48,143:PRINT "H":LO
CATE48,152:PRINT "I":LOCATE 48,161:PRINT
 "N":LOCATE 48,170:PRINT "G"
25 PSET(40,75):DRAW"U3L3D3R3E1U3L3G1BR18
0BU16E3F3L4C15BU25BL70L10L10U20R20D20G10
L20E10U20G10D20R20U20E10G10L20"
26 LOCATE 0,184:COLOR 5:PRINTSTRING$(43,
201):LINE (0,184)-(256,184),1
27 FOR S=0 TO 1:S$="":FOR T=1 TO 32:READ
 A$:S$=S$+CHR$(VAL("&h"+A$)):NEXT T:SPRI
TE$(S)=S$:NEXT S
28 FOR S=3 TO 5:S$="":FOR T=1 TO 8:READ
A$:S$=S$+CHR$(VAL("&h"+A$)):NEXT T:SPRIT
E$(S)=S$:NEXT S
29 Y=96: X=128
30 ' ***          Main loop          ***
31 IF STRIG(1) OR STRIG(0) THEN GOTO 84
32 UP=INT(RND(1)*180)
33 DOWN=INT(RND(1)*240)
34 COUNT=CO+1
35 IF ST=0 AND CO>CU THEN CU=CU-1:BEEP
36 IF CO>CU OR STR=1 THEN PUT SPRITE 3,(
DOWN+4,UP+4),9,3:PUT SPRITE 4,(DOWN+4,UP
+1),3,4:STR=0:CO=0:BEEP:CQ=CQ+1
37 IF CU<0 THEN GOTO 68
38 IF CU>CW THEN CW=INT(RND(1)*HE):C3=0:
CW=CW+CW
39 IF CU=<CW AND C3=0 THEN LOCATE 226,96
:COLOR 14:PRINT STRING$(2,201):LOCATE 22
6,105:PRINTSTRING$(2,201):C3=1
40 ' *** Empty strawberries ***
41 IF C3=1 AND CY>0 AND POINT (X+8,Y+12)
=14 THEN CE=0:FOR CR=1 TO CY:CE=CE+CR+2:
NEXT CR:SC=SC+CE:BEEP:BEEP:BEEP:BEEP:Y4=
0:CY=-1
42 IF CQ>=CW AND C3=1 THEN LOCATE 226,96
:COLOR 4:PRINT STRING$(2,201):LOCATE 226
,105:PRINT STRING$(2,201)
43 TI=TI+.2
44 ' *** Test for edge of screen ***
45 X=X+X1:IF X1<0 THEN X1=X1+.09 ELSE IF
 X1>0 THEN X1=X1-.09
46 IF X<5 THEN X=5:X1=0:ELSE IF X>235 TH
EN X=235:X1=0
47 IF X1<4 AND X1>-4 THEN Y1=Y1+.01
48 Y=Y+Y1+Y4
49 IF Y<0 THEN Y=0 ELSE IF Y>168 THEN Y=
168:GOTO90
50 PUTSPRITE0,(X,Y),15,SP
51 IF SC>3000 THEN DS=1
52 SY=0
53 ' *** Stick routines ***
54 D=STICK(1) OR STICK(0)
55 ON D+1 GOTO64,56,57,58,59,60,61,62,63
56 PUT SPRITE 0,(X,Y),15,SP
57 Y=Y-6:Y1=Y1-.9:X1=X1+.8:SP=0:SO=1:GOT
O 64
58 X1=X1+1.1:SP=0:SO=1:GOTO 64
59 Y=Y+6:Y1=Y1+.8:X1=X1+.8:SP=0:SO=1:GOT
O 64
60 Y=Y+7:Y1=Y1+.9:GOTO64
61 Y=Y+6:Y1=Y1+.8:X1=X1-.8:SP=1:SO=1:GOT
O 64
62 X1=X1-1.1:SP=1:SO=1:GOTO 64
63 Y=Y-6:Y1=Y1-.9:X1=X1-.8:SP=1:SO=1
64 IF Y1<0 THEN Y1=0
65 GOTO 31
66 ' ***      END OF FRAME       ***
67 ' *** PRINT SCORE AND BONUS ***
68 COLOR 15:SCREEN 0,0:LOCATE 5,5:PRINT
"YOUR TIME =";INT(TI):LOCATE 5,10:PRINT
"MAX. TIME FOR BONUS =";B1:LOCATE5,15:PR
INT "GOOD!!!":BSC=B1-TI+1:FOR T=0 TO BSC
:LOCATE 5,20:PRINT "BONUS ";T*100;  ":sc
ore = " SC+T*100:B=T*100:BEEP:NEXT T
69 IF B<1 THEN GOTO 76
70 ' *** Bonus tune ***
71 PLAY"O2T255L20CCDDEECDCCDDEEDDCCDDEEF
FEEFFGGG"
72 PLAY"R4804T255L20CCDDEECDCCDDEEDDCCDD
EEFFEEFFGGGG"
73 PLAY"T25502L20EEDDCCDEDDCCDDEDCCDDEED
EDDEEDEDCDEDCCCC","T255L2004EEDDCCDEDDCC
DDEDCCDDEEDEDDEEDEDCDEDCCCC"
74 LOCATE 3,25:PRINT "C O N G R A T U L
A T I O N S ! ! !"
75 ' *** Reset variables for next screen
 ***
76 STR=1:RN=RN-3:Y4=0:CY=0:C3=0:B=0:TI=0
:HE=HE-HE*.03:X1=0:Y1=0:CW=0
77 FOR T=1 TO 1500:NEXT:RESTORE:GOTO 20
78 FOR T=1 TO 5500:NEXT:RESTORE:GOTO 20
79 ' *** Pick up strawberry ***
80 FOR S=3 TO 4:PUT SPRITE S,(-10,-10),4
,S:NEXT S:SC=SC+20:CU=CU-1:STR=1
81 CY=CY+1:Y4=Y4+.2
82 RETURN
83 ' *** Extra thrust (fire button) ***
84 IF SP=0 THEN X1=X1+.4 ELSE IF SP=1 TH
EN X1=X1-.4
85 GOTO 32
86 PUT SPRITE 5,(X+X4,Y+18),15,5
87 GOTO 31
88 '**       COLLISION ROUTINE        **
89 '** and print score  & high score **
```

```
90 PUT SPRITE 0,(-10,-10),4,SP:LOCATE X,
Y+6:FOR T=1 TO 20:V=V-.5:COLOR6:PRINT "*
":COLOR 10:LOCATE X,Y+6:PRINT CHR$(189):
COLOR 6:LOCATE X,Y+6:PRINT CHR$(211):LOC
ATE X,Y+6:PRINT CHR$(172):SOUND 7,&B1111
0111:SOUND 6,20:SOUND 8,V:NEXTT:SOUND 7,
&B11111110
91 SOUND6,0:COLOR 15,4,5
92 LOCATE0,45:PRINT"          YOU SCORED
: ";SC
93 IFSC>HSTHENHS=SC
94 LOCATE0,60:PRINT"          HI-SCORE :
";HS
95 LOCATE0,100:PRINT"          PRESS FIRE
 TO PLAY AGAIN"
96 IFSTRIG(1)ORSTRIG(0)THENGOTO16
97 GOTO96
98 ' ***        SPRITE         ***
99 ' ***        VALUES         ***
100 DATA 00,00,00,00,00,00
101 DATA 80,C0,E0,F8,FF,FF
102 DATA 8A,8A,FF,FF,00,00
103 DATA 00,00,00,00,00,00
104 DATA 00,00,F8,FC,0A,0B
105 DATA FF,FE,00,00,00,00
106 DATA 00,00,00,00,00,00
107 DATA 1F,3F,50,D0,FF,7F
108 DATA 00,00,00,00,00,00
109 DATA 01,03,07,1F,FF,FF
110 DATA 51,51,FF,FF,18,76
111 DATA DB,AD,D7,6E,3C,18
112 DATA A5,E7,44,00,00,00
113 DATA 00,00,00,00,00,00
114 DATA FF,FF,00,00,00,00
115 ' ***       INSTRUCTIONS       ***
116 CLS
117 LOCATE 0,2:PRINT "There are a whole
mob of crazy mutant    strawberries just
'hanging around'       polluting the airs
pace.  Your job is to capture these muta
nts in your deadly     little aircraft an
d put them in the      fermentation chamb
er."
118 PRINT
119 PRINT "To do this you manuevour your
 aircraft  around the landscape and fly
through thestrawberries to catch them."
120 PRINT
121 PRINT "The more strawberries you cat
ch, the    heavier your ship gets (and i
t becomes  very sluggish!) and so it mus
t be empt- ied in the fermentation chamb
er (repres-ented as two silver squares).
"
122 PRINT
123 PRINT:PRINT:PRINT "Press any key to
continue."
124 IF INKEY$="" THEN 124
125 CLS:LOCATE 0,4
126 PRINT "The more strawberries you unl
oad at     once, the higher the score."
127 PRINT
128 PRINT"If you cross the black line at
 the      bottom of the screen you will
be blown  to pieces."
129 PRINT
130 PRINT:PRINT "                GOOD LUCK!
!!"
131 PRINT:PRINT:PRINT "Press any key to
begin."
132 IF INKEY$="" THEN 132
133 RETURN
```

NOTE: MSX owners change or add the following lines -

```
7 COLOR 15,4,5:SCREEN 0,0:COLOR 10,4,3:L
OCATE2,5:PRINT"Strawberry":LOCATE2,6:PRI
NT"Catcher":FOR T=1 TO 1000:NEXT
11 CLEAR 500:OPEN"GRP:"AS#1
13 COLOR 10,4,3:LOCATE 2,5:PRINT"Strawbe
rry":LOCATE2,6:PRINT"Catcher"
21 SCREEN 1,2:COLOR15,4,3:CU=50:TI=0
22 FOR Y=55 TO 196 STEP 2:LINE (0,Y)-(25
6,Y),12:C=C+1:Y=Y+C:NEXT:LINE (0,55)-(25
6,55),7:CIRCLE (250,0),30,10,3.14,6.28:P
AINT(250,0),10:PRESET(222,9):COLOR1:PRIN
T#1,"  SUN"
24 PRESET(132,112):PRINT#1,"CUBE":PRESET
(48,134):PRINT#1,"T":PRESET(48,143):PRIN
T#1,"H":PRESET(48,152):PRINT#1,"I":PRESE
T(48,161):PRINT#1,"N":PRESET(48,170):PRI
NT#1,"G"
26 PRESET(0,184):COLOR 5:PRINT#1,STRING$
(43,219):LINE (0,184)-(256,184),1
39 IF CU=<CW AND C3=0 THEN PRESET(226,96
):COLOR 14:PRINT#1,STRING$(2,219):PRESET
(226,105):PRINT#1,STRING$(2,219):C3=1
42 IF CQ>=CW AND C3=1 THEN PRESET(226,96
):COLOR 4:PRINT#1,STRING$(2,219):PRESET(
226,105):PRINT#1,STRING$(2,219)
90 PUT SPRITE 0,(-10,-10),4,SP:PRESET(X,
Y+6):FOR T=1 TO 20:V=V-.5:COLOR6:PRINT#1
,"*":COLOR 10:LOCATE(X,Y+6):PRINT#1,CHR$
(28):COLOR 6:PRESET(X,Y+6):PRINT#1,CHR$(
22):PRESET(X,Y+6):PRINT#1,CHR$(23):SOUND
 7,&B11110111:SOUND 6,20:SOUND 8,V:NEXTT
:SOUND 7,&B111
92 PRESET(0,45):PRINT#1,"      YOU SCORED
 : ";SC
94 PRESET(0,60):PRINT#1,"      HI-SCORE :
 ";HS
95 PRESET(0,100):PRINT#1,"      PRESS FIR
E TO PLAY AGAIN"
```

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